Lecture 10: Covariance and Correlation, Conditional Expectation and Variance

Jamie Haddock

Table of contents

1 Covariance and Correlation **1**
1.1 Covariance . . . 1
1.1.1 Example: Uncorrelated but Not Independent . . . 1
1.2 Correlation . . . 2
1.2.1 Example: Heads and Tails out of n Flips . . . 2
1.3 Variance of a Sum of RVs . . . 3
1.3.1 Example: The Hat-Check Problem, Revisited . . . 3

2 Conditional Expectation & Variance **3**
2.1 Law of Iterated Expectation . . . 3
2.1.1 Example: Random Coin Bias . . . 3
2.1.2 Example: Averaging Quiz Scores by Section . . . 4
2.2 Conditional Expectation as an Estimator . . . 4
2.3 Conditional Variance . . . 5
2.3.1 Example: Biased-Coin Example, Variance . . . 6

1 Covariance and Correlation

1.1 Covariance

$$\mathrm{Cov}(X,Y) = E[(X-E[X])(Y-E[Y])] = E[XY] - E[X]E[Y]$$

Definition: uncorrelated

X and Y are **uncorrelated** if $\mathrm{Cov}(X,Y) = 0$.

Facts

- $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$
- $\mathrm{Cov}(X, aY+b) = a\,\mathrm{Cov}(X,Y)$
- $\mathrm{Cov}(X, Y+Z) = \mathrm{Cov}(X,Y) + \mathrm{Cov}(X,Z)$
- X, Y independent $\implies \mathrm{Cov}(X,Y) = 0$ (the converse is **false** – next slide)

1.1.1 Example: Uncorrelated but Not Independent

(X,Y) uniform over the four points $(1,0),(0,1),(-1,0),(0,-1)$ (each with probability $\frac{1}{4}$)

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y] = 0 - 0\dot{0} = 0$

However, X and Y are not independent since a nonzero value of X fixes $Y = 0$ and vice versa.

1.2 Correlation

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} \in [-1,1]$$

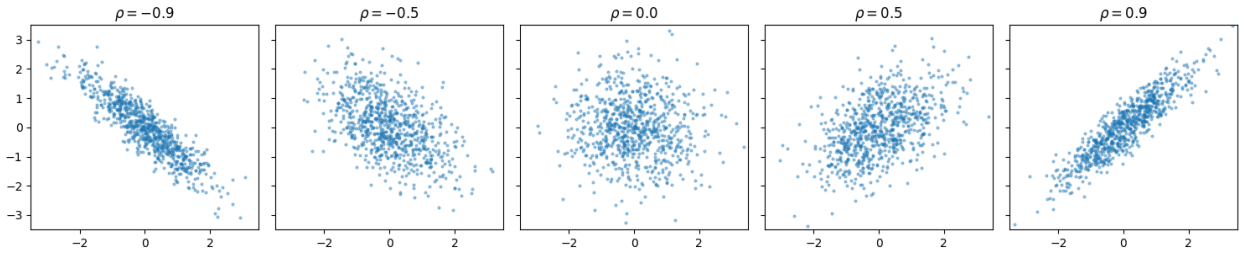
ρ rescales covariance to be unit-free: ρ near $+1$ means a strong positive linear relationship, near -1 a strong negative one, near 0 little to no linear relationship.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1)
rhos = [-0.9, -0.5, 0.0, 0.5, 0.9]

fig, axes = plt.subplots(1, len(rhos), figsize=(15, 3.2), sharex=True, sharey=True)
for ax, r in zip(axes, rhos):
    cov_mat = [[1, r], [r, 1]]
    samples = rng.multivariate_normal([0, 0], cov_mat, 800)
    ax.scatter(samples[:, 0], samples[:, 1], s=4, alpha=0.4, color="C0")
    ax.set_title(fr"$\rho = {r}$")
    ax.set_xlim(-3.5, 3.5)
    ax.set_ylim(-3.5, 3.5)

plt.tight_layout()
plt.show()
```



1.2.1 Example: Heads and Tails out of n Flips

$X = \#$ heads, $Y = \#$ tails out of n independent flips. Since $X + Y = n$ and $E[X] + E[Y] = n$, we have

$$X - E[X] = -(Y - E[Y]).$$

Thus,

$$\text{Cov}(X,Y) = E\left[(X - E[X])(Y - E[Y])\right] = -E\left[(X - E[X])^2\right] = -\text{var}(X).$$

And,

$$\rho(X,Y) = \frac{-\text{var}(X)}{\sqrt{\text{var}(X)^2}} = -1.$$

> **Fact**
>
> $\rho = 1$ or $-1 \iff Y - E[Y] = c\,(X - E[X])$ for some constant c (a perfect linear relationship). Here $c = -1$.

1.3 Variance of a Sum of RVs

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + \sum_{i\neq j} \mathrm{Cov}(X_i, X_j)$$

Proof

$$\mathrm{Var}\Big(\sum_i X_i\Big) = E\Big[\Big(\sum_i (X_i - EX_i)\Big)^2\Big] = \sum_i \sum_j E[(X_i - EX_i)(X_j - EX_j)]$$

Splitting the double sum into $i = j$ and $i \neq j$ terms gives $\sum_i \mathrm{Var}(X_i) + \sum_{i\neq j} \mathrm{Cov}(X_i, X_j)$. (The second sum is over *ordered* pairs, so it's the same as $2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$.)

If all pairs are uncorrelated (in particular if the X_i are independent), every covariance term vanishes and this collapses to the familiar $\mathrm{Var}(\sum_i X_i) = \sum_i \mathrm{Var}(X_i)$.

1.3.1 Example: The Hat-Check Problem, Revisited

Recall the previous example: n people's hats are shuffled and returned at random; X = number of people who get their own hat back, $X = \sum_i X_i$ where X_i indicates person i getting their own hat. We showed $E[X] = 1$ for every n. Now: $\mathrm{Var}(X)$?

Because X_i is Bernoulli with parameter $p = 1/n$, we have

$$\mathrm{Var}(X_i) = \tfrac{1}{n} - \tfrac{1}{n^2} = \tfrac{n-1}{n^2} \text{ and } E[X_i] = 1/n,$$

so

$$\begin{aligned}\mathrm{Cov}(X_i, X_j) &= E[X_i X_j] - \frac{1}{n}\frac{1}{n} = P(X_i = 1, X_j = 1) - \frac{1}{n^2} \\ &= P(X_i = 1)P(X_j = 1 \mid X_i = 1) - \frac{1}{n^2} = \tfrac{1}{n(n-1)} - \tfrac{1}{n^2} = \tfrac{1}{n^2(n-1)} \quad (i \neq j)\end{aligned}$$

Thus,

$$\mathrm{Var}(X) = n \cdot \frac{n-1}{n^2} + n(n-1) \cdot \frac{1}{n^2(n-1)} = \frac{n-1}{n} + \frac{1}{n} = 1$$

Var$(X) = 1$ **for every** $n \geq 2$.

2 Conditional Expectation & Variance

2.1 Law of Iterated Expectation

$E[X \mid Y]$ is itself a **random variable**: its value depends on the outcome of Y. $E[X \mid Y{=}y]$ is an ordinary function of y; plugging in the random Y makes $E[X \mid Y]$ random too.

> Law of Iterated Expectation (LIE)
>
> $$E[E[X \mid Y]] = E[X]$$

2.1.1 Example: Random Coin Bias

A coin's probability of heads Y is itself random ($Y \sim \mathrm{Uniform}(0, 1)$). Given $Y = y$, the number of heads X in n flips is Binomial(n, y). What is $E[X]$?

$$E[X \mid Y] = nY \quad \Longrightarrow \quad E[X] = E[nY] = n\,E[Y] = \frac{n}{2}$$

2.1.2 Example: Averaging Quiz Scores by Section

A class has n students and the ith student's quiz score is x_i, with the class's average

$$m = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The students are divided into k disjoint sections $A_1, \cdots, A_k$ and section A_i has n_i students. The average score in section i is

$$m_i = \frac{1}{n_i}\sum_{i \in A_i} x_i.$$

The average over the whole class can be computed as a weighted average of the section averages:

$$\sum_{s=1}^{k} \frac{n_s}{n} m_s = \sum_{s=1}^{k} \frac{n_s}{n} \frac{1}{n_s} \sum_{i \in A_s} x_i = \frac{1}{n} \sum_{s=1}^{k} \sum_{i \in A_s} x_i = m.$$

We can see this same calculation through the conditional expectation lens:

Consider an experiment where a student is selected uniformly at random. Consider X to be the quiz score of the student, and Y to be the section of the student.

We have $E[X] = m$ and $E[X \mid Y = s] = m_s$, then

$$m = E[X] = E[E[X \mid Y]] = \sum_{s=1}^{k} E[X \mid Y = s]P(Y = s) = \sum_{s=1}^{k} \frac{n_s}{n} m_s.$$

2.2 Conditional Expectation as an Estimator

Write $\hat{X} = E[X \mid Y]$ (the estimator) and $\tilde{X} = X - \hat{X}$ (the estimation error), so $X = \hat{X} + \tilde{X}$.

> Facts
>
> - **Estimator is unbiased:** $E[\tilde{X}] = E[X] - E[\hat{X}] = E[X] - E[X] = 0$ (by LIE).
> - **Uncorrelated with the error:** $\mathrm{Cov}(\hat{X}, \tilde{X}) = 0$, since conditioning on $Y = y$ makes $\hat{X}$ a constant and $E[\tilde{X} \mid Y{=}y] = 0$.
> - Consequently $\mathrm{Var}(X) = \mathrm{Var}(\hat{X}) + \mathrm{Var}(\tilde{X})$.

The picture: $\hat{X}(y)$ is the conditional-mean curve; $\tilde{X}$ is the scatter of points around it, and that scatter is – on average – unrelated to where you are on the curve.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(6)

def draw(n):
    Y = rng.uniform(0, 1, n)
    Xhat = Y**2                                  # conditional mean E[X|Y=y] = y^2
    X = rng.normal(Xhat, 0.08, n)                # X | Y=y ~ N(y^2, 0.08^2)
    return Y, Xhat, X, X - Xhat

# a modest sample for a readable scatter plot...
```

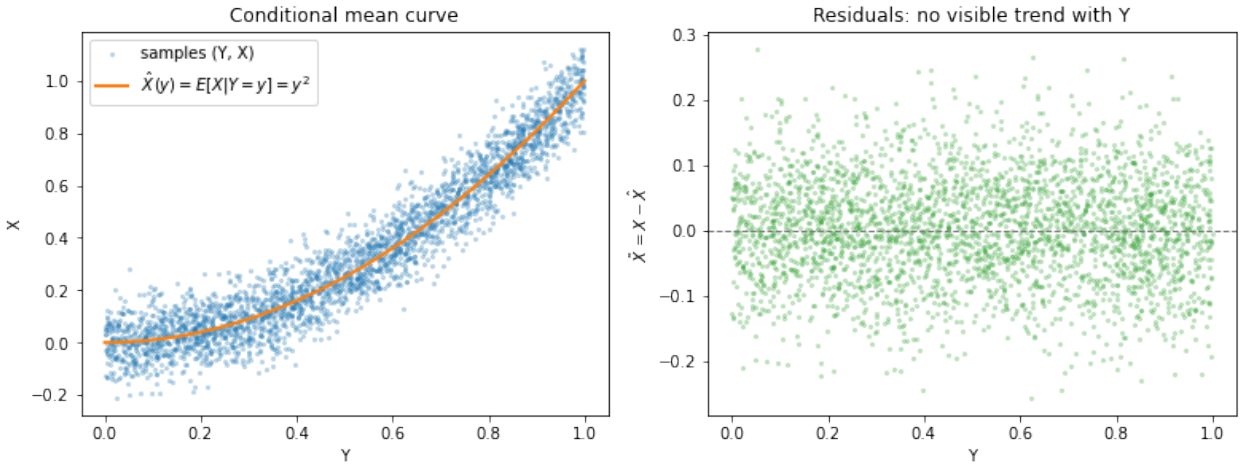
```
Y, Xhat, X, Xtilde = draw(3000)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.3))

yy = np.linspace(0, 1, 200)
axes[0].scatter(Y, X, s=4, alpha=0.25, color="C0", label="samples (Y, X)")
axes[0].plot(yy, yy**2, color="C1", lw=2, label=r"$\hat X(y) = E[X|Y=y] = y^2$")
axes[0].set_xlabel("Y")
axes[0].set_ylabel("X")
axes[0].legend()
axes[0].set_title("Conditional mean curve")

axes[1].scatter(Y, Xtilde, s=4, alpha=0.25, color="C2")
axes[1].axhline(0, color="gray", ls="--", lw=1)
axes[1].set_xlabel("Y")
axes[1].set_ylabel(r"$\tilde X = X - \hat X$")
axes[1].set_title("Residuals: no visible trend with Y")

plt.tight_layout()
plt.show()
```



2.3 Conditional Variance

We now define the random variable

$$\mathrm{Var}(X \mid Y) = E\left[(X - E[X \mid Y])^2 \mid Y\right] = E[\tilde{X}^2 \mid Y].$$

Law of Total Variance

$$\mathrm{Var}(X) = E\big[\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(E[X \mid Y]\big)$$

Derivation

This is exactly the previous slide's decomposition $\mathrm{Var}(X) = \mathrm{Var}(\hat{X}) + \mathrm{Var}(\tilde{X})$, rewritten: the "estimator variance" term $\mathrm{Var}(\hat{X}) = \mathrm{Var}(E[X|Y])$ is the **between-group** variance, and the "error variance" term $\mathrm{Var}(\tilde{X}) = E[\mathrm{Var}(X|Y)]$ is the **within-group** variance, since $E[\tilde{X}^2 \mid Y{=}y] = \mathrm{Var}(X \mid Y{=}y)$ (the conditional mean of $\tilde{X}$ given $Y = y$ is 0).

2.3.1 Example: Biased-Coin Example, Variance

Continuing the random-bias coin: with $Y \sim \text{Uniform}(0,1)$, $X \mid Y = y \sim \text{Binomial}(n, y)$,

$$\begin{aligned} E[\text{Var}(X|Y)] &= E[nY(1-Y)] = n(E[Y] - \text{Var}(Y) - (E[Y])^2) = n\Big(\tfrac{1}{2} - \tfrac{1}{12} - \tfrac{1}{4}\Big) \\ &= \frac{n}{6} \end{aligned}$$

Then,

$$\text{Var}(E[X|Y]) = \text{Var}(nY) = n^2 \cdot \tfrac{1}{12}$$

and,

$$\text{Var}(X) = \frac{n}{6} + \frac{n^2}{12} = \frac{n(n+2)}{12}.$$